



3.3.5 Forming Q

03.3.5 Forming Q, part 1

fit width



Given $A \in \mathbb{C}^{m \times n}$, let $[A, t] = \text{HQR_unb_var1}(A)$ yield the matrix A with the Householder vectors stored below the diagonal, R stored on and above the diagonal, and the scalars τ_i , $0 \leq i < n$, stored in vector t . We now discuss how to form the first n columns of $Q = H_0 H_1 \cdots H_{n-1}$. The computation is illustrated in [Figure 3.3.5.1](#).

Original matrix	$\left(\begin{array}{c c} \alpha_{11} & a_{12}^T \\ a_{21} & A_{22} \end{array} \right) :=$	“Move forward”
	$\left(\begin{array}{c c} 1 - 1/\tau_1 & -(u_{21}^H A_{22})/\tau_1 \\ -u_{21}/\tau_1 & A_{22} + u_{21} a_{12}^T \end{array} \right)$	
$\begin{array}{c c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 0 \end{array}$	$\begin{array}{c c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \times \\ \hline 0 & 0 & 0 & \times \end{array}$	$\begin{array}{c c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & \times \\ 0 & 0 & 0 & \times \end{array}$
	$\begin{array}{c c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \times & \times \\ \hline 0 & 0 & \times & \times \\ 0 & 0 & \times & \times \end{array}$	$\begin{array}{c c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & \times & \times \\ 0 & 0 & \times & \times \\ 0 & 0 & \times & \times \end{array}$
	$\begin{array}{c c} 1 & 0 & 0 & 0 \\ 0 & \times & \times & \times \\ \hline 0 & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & \times & \times & \times \end{array}$	$\begin{array}{c c} 1 & 0 & 0 & 0 \\ 0 & \times & \times & \times \\ \hline 0 & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & \times & \times & \times \end{array}$
	$\begin{array}{c c} \times & \times & \times & \times \\ \hline \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \end{array}$	$\begin{array}{c c} \times & \times & \times & \times \\ \hline \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \end{array}$

Figure 3.3.5.1. Illustration of the computation of Q .

Notice that to pick out the first n columns we must form

$$\begin{aligned}
 & Q \left(\begin{array}{c} I_{n \times n} \\ 0 \end{array} \right) \\
 &= \\
 & H_0 \cdots H_{n-1} \left(\begin{array}{c} I_{n \times n} \\ 0 \end{array} \right) \\
 &= \\
 & \underbrace{H_0 \cdots H_{k-1} H_k \cdots H_{n-1}}_{B_k} \left(\begin{array}{c} I_{n \times n} \\ 0 \end{array} \right)
 \end{aligned}$$

so that $Q = B_0$, where $B_k = H_k \cdots H_{n-1} \left(\begin{array}{c} I_{n \times n} \\ 0 \end{array} \right)$.

Homework 3.3.5.1. ALWAYS/SOMETIMES/NEVER:

$$B_k = H_k \cdots H_{n-1} \left(\begin{array}{c} I_{n \times n} \\ 0 \end{array} \right) = \left(\begin{array}{c|c} I_{k \times k} & 0 \\ \hline 0 & \tilde{B}_k \end{array} \right).$$

for some $(m-k) \times (n-k)$ matrix $\tilde{B}_k$.

[Answer](#) [Solution](#)

The proof of this is by induction on k :

- Base case: $k = n$. Then $B_n = \left(\begin{array}{c} I_{n \times n} \\ 0 \end{array} \right)$, which has the desired form.
- Inductive step: Assume the result is true for B_k . We show it is true for B_{k-1} :

$$\begin{aligned}
& B_{k-1} \\
& = \\
& H_{k-1} H_k \cdots H_{n-1} \left(\frac{I_{n \times n}}{0} \right) \\
& = \\
& H_{k-1} B_k \\
& = \\
& H_{k-1} \left(\frac{I_{k \times k}}{0} \left| \frac{0}{\tilde{B}_k} \right. \right) \\
& = \\
& \left(\begin{array}{c|c} I_{(k-1) \times (k-1)} & 0 \\ \hline 0 & I - \frac{1}{\tau_k} \left(\frac{1}{u_k} \right) \left(1 \mid u_k^H \right) \end{array} \right) \left(\begin{array}{c|c} I_{(k-1) \times (k-1)} & 0 \\ \hline 0 & 1 \end{array} \left| \frac{0}{\tilde{B}_k} \right. \right) \\
& = \\
& \left(\begin{array}{c|c} I_{(k-1) \times (k-1)} & 0 \\ \hline 0 & \left(I - \frac{1}{\tau_k} \left(\frac{1}{u_k} \right) \left(1 \mid u_k^H \right) \right) \left(\frac{1}{0} \left| \frac{0}{\tilde{B}_k} \right. \right) \end{array} \right) \\
& = < \text{choose } y_k^T = u_k^H \tilde{B}_k / \tau_k > \\
& \left(\begin{array}{c|c} I_{(k-1) \times (k-1)} & 0 \\ \hline 0 & \left(\frac{1}{0} \left| \frac{0}{\tilde{B}_k} \right. \right) - \left(\frac{1}{u_k} \right) \left(1 / \tau_k \mid y_k^T \right) \end{array} \right) \\
& = \\
& \left(\begin{array}{c|c} I_{(k-1) \times (k-1)} & 0 \\ \hline 0 & \left(\frac{1 - 1/\tau_k}{-u_k/\tau_k} \left| \frac{-y_k^T}{\tilde{B}_k - u_k y_k^T} \right. \right) \end{array} \right) \\
& = \\
& \left(\begin{array}{cc|c} I_{(k-1) \times (k-1)} & 0 & 0 \\ \hline 0 & 1 - 1/\tau_k & -y_k^T \\ \hline 0 & -u_k/\tau_k & \tilde{B}_k - u_k y_k^T \end{array} \right) \\
& = \\
& \left(\begin{array}{c|c} I_{(k-1) \times (k-1)} & 0 \\ \hline 0 & \tilde{B}_{k-1} \end{array} \right).
\end{aligned}$$

- By the Principle of Mathematical Induction the result holds for $B_0, \dots, B_n$.

ALWAYS

03.3.5 Forming Q, part 2

fit width



The last exercise justifies the algorithm in [Figure 3.3.5.2](#).

$[A] = \text{FormQ}(A, t)$

$A \rightarrow \left(\frac{A_{TL}}{A_{BL}} \mid \frac{A_{TR}}{A_{BR}} \right), t \rightarrow \left(\frac{t_T}{t_B} \right)$

A_{TL} is $n(A) \times n(A)$ and t_T has $n(A)$ elements

while $n(A_{TL}) > 0$

$\left(\frac{A_{TL}}{A_{BL}} \mid \frac{A_{TR}}{A_{BR}} \right) \rightarrow \left(\frac{A_{00}}{a_{10}^T} \mid \frac{a_{01}}{\alpha_{11}} \mid \frac{A_{02}}{a_{12}^T} \right), \left(\frac{t_T}{t_B} \right) \rightarrow \left(\frac{t_0}{\tau_1} \right)$

Update $\left(\frac{\alpha_{11}}{a_{21}} \mid \frac{a_{12}^T}{A_{22}} \right) :=$

$\left(I - \frac{1}{\tau_1} \left(\frac{1}{u_{21}} \right) \left(1 \mid u_{21}^H \right) \right) \left(\frac{1}{0} \mid \frac{0}{A_{22}} \right)$

via the steps

$\alpha_{11} := 1 - 1/\tau_1$

$a_{12}^T := -(a_{21}^H A_{22})/\tau_1$

$A_{22} := A_{22} + a_{21} a_{12}^T$

$a_{21} := -a_{21}/\tau_1$

$\left(\frac{A_{TL}}{A_{BL}} \mid \frac{A_{TR}}{A_{BR}} \right) \leftarrow \left(\frac{A_{00}}{a_{10}^T} \mid \frac{a_{01}}{\alpha_{11}} \mid \frac{A_{02}}{a_{12}^T} \right), \left(\frac{t_T}{t_B} \right) \leftarrow \left(\frac{t_0}{\tau_1} \right)$

endwhile

Figure 3.3.5.2. Algorithm for overwriting A with Q from the Householder transformations stored as Householder vectors below the diagonal of A (as produced

by $[A, t] = \text{HQR_unb_var1}(A, t)$),
 which, given $[A, t] = \text{HQR_unb_var1}(A)$ from [Figure 3.3.4.2](#), overwrites A with
 the first $n = n(A)$ columns of Q .

Homework 3.3.5.2. Implement the algorithm in [Figure 3.3.5.2](#) as

```
function [ A_out ] = FormQ( A, t )
```

by completing the code in [Assignments/Week03/matlab/FormQ.m](#). You will want to use [Assignments/Week03/matlab/test_FormQ.m](#) to check your implementation. Input is the $m \times n$ matrix A and vector t that resulted from $[A, t] = \text{HQR}(A)$. Output is the matrix Q for the QR factorization. You may want to use [Assignments/Week03/matlab/test_FormQ.m](#) to check your implementation.

▼ [Solution](#)

See [Assignments/Week03/answers/FormQ.m](#)

Homework 3.3.5.3. Given $A \in \mathbb{C}^{m \times n}$, show that the cost of the algorithm in [Figure 3.3.5.2](#) is given by

$$C_{\text{FormQ}}(m, n) \approx 2mn^2 - \frac{2}{3}n^3 \text{ flops.}$$

▼ [Hint](#) ▼ [Solution](#)

Modify the answer for [Homework 3.3.4.2](#).

When computing the Householder QR factorization, the bulk of the cost is in the computations

$$w_{12}^T := (a_{12}^T + u_{21}^H A_{22})/\tau_1$$

and

$$A_{22} - u_{21} w_{12}^T.$$

When forming Q , the cost is in computing

$$a_{12}^T := -(a_{21}^H A_{22})/\tau_1$$

and

$$A_{22} := A_{22} + u_{21} w_{12}^T.$$

During the iteration when A_{TL} is $k \times k$, these represent, essentially, identical costs: the matrix-vector multiplication $(u_{21}^H A_{22})$ and rank-1 update with matrix A_{22} which is of size approximately $(m - k) \times (n - k)$ for a cost of $4(m - k)(n - k)$ flops. Thus the total cost is approximately

$$\begin{aligned} &\sum_{k=n-1}^0 4(m - k)(n - k) \\ &= \text{< reverse the order of the summation >} \\ &\sum_{k=0}^{n-1} 4(m - k)(n - k) \\ &= \\ &4 \sum_{j=1}^n (m - n + j)j \\ &= \\ &4(m - n) \sum_{j=1}^n j + 4 \sum_{j=1}^n j^2 \\ &= \\ &2(m - n)n(n + 1) + 4 \sum_{j=1}^n j^2 \\ &\approx \\ &2(m - n)n^2 + 4 \int_0^n x^2 dx \\ &= \\ &2mn^2 - 2n^3 + \frac{4}{3}n^3 \\ &= \\ &2mn^2 - \frac{2}{3}n^3. \end{aligned}$$

Ponder This 3.3.5.4. If $m = n$ then Q could be accumulated by the sequence

$$Q = (\cdots((IH_0)H_1) \cdots H_{n-1}).$$

Give a high-level reason why this would be (much) more expensive than the algorithm in [Figure 3.3.5.2](#)